Sukrut Shishupal, PhD

Biomedical Informatics | Time Series Machine Learning | Research Software

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RESEARCH PROFILE

Researcher developing reusable temporal representations and software for environmental and healthcare data. Built a library of 64.8 million exposure patterns from U.S. EPA monitoring records and a deployed application for searching and downloading them. Experience spans Python pipelines, HPC execution, geospatial cohort linkage, and exploratory analysis of clinical and EHR activity data.

RESEARCH AND SOFTWARE EXPERIENCE

Research Associate | University of Utah, Biomedical Informatics

Current appointment

Graduate Research Assistant | University of Utah

November 2022 – October 2026

- Built the SMARTER exposure-pattern library from raw EPA monitoring records through a reusable research resource and first-author IEEE ICHI 2026 publication.
- Implemented Python extraction and quality-control pipelines executed as SLURM jobs on the CHPC cluster, addressing missing observations, uneven temporal coverage, and redundant patterns.
- Developed and deployed a Flask application using Docker and Nomad to search, filter, and download patterns by location and time: https://ehie-shapelets.ctsi.utah.edu.
- Linked clinical records to environmental monitoring data by geographic proximity and prediagnosis time windows for an exploratory hypersensitivity pneumonitis study.
- Produced analyses and visualizations for published telemedicine studies of access disparities and travel-related emissions savings.

Trainee Software Developer | Automation Teknix, India

November 2020 – May 2021

- Developed software for industrial automation projects.

TECHNICAL SKILLS

Programming and analysis: Python, NumPy, pandas, scikit-learn, R, SQL, Java

Methods: Time-series representations, shapelet methods, feature engineering, clustering, statistical modeling, cross-validation, geospatial analysis

Research software and computing: Flask, Docker, Nomad, HPC, SLURM, data visualization

SELECTED RESEARCH PROJECTS

Reusable environmental exposure patterns

- Extracted hourly and daily patterns in 7- and 30-day windows from EPA monitoring records spanning 2004–2024. Organized the library for reuse across studies without repeating extraction from raw observations.
- Applied observation-completeness criteria, bounded gap interpolation, site-level normalization, and quality-based window selection to construct the pattern collection.

Hypersensitivity pneumonitis and air-pollution exposure

- Matched a clinical cohort to nearby monitoring sites and characterized exposure during the two years before diagnosis. Compared pattern-based clusters across fibrotic and non-fibrotic groups in an exploratory analysis.

Nursing EHR activity patterns

- Represented within-shift EHR interactions in 5-minute bins and extracted overlapping 30-minute windows to study local activity patterns. Used clustering to characterize recurring temporal behavior in ICU nursing workflows.

Protein binding affinity

- Compared graph-based representations of 101 heterodimeric protein complexes as features for machine learning models; evaluated models using cross-validation.

EDUCATION

PhD, Biomedical Informatics | University of Utah | 2026

Dissertation focus: local temporal-pattern representations for environmental and healthcare data.

MS, Biomedical Informatics | University of Utah | 2024

Integrated BS–MS, Bioinformatics and Biotechnology

Savitribai Phule Pune University, India | 2020

SELECTED PUBLICATIONS

Shishupal S, et al. A reusable library of exposure health machine learning primitives. IEEE International Conference on Healthcare Informatics. 2026:481–489. doi:10.1109/ICHI69079.2026.00272. First author.

Social vulnerability, lower broadband internet access, and rurality associated with lower telemedicine use in U.S. counties. JAMIA Open. 2025. doi:10.1093/jamiaopen/ooaf056. Co-author.

Travel distance between participants in US telemedicine sessions with estimates of emissions savings: observational study. Journal of Medical Internet Research. 2024. doi:10.2196/53437. Second author.

TEACHING

Graduate Teaching Assistant | University of Utah | August 2023 – October 2026

Co-developed course materials and taught sessions for Introduction to Programming (Fall 2023) and Biomedical Data Wrangling (Spring 2024 and Spring 2026).